

MFE 230G: Homework 1

Due: August 24, 2025

1. (20 points) Consider the following two-stock universe. Stock A has volatility 30%. Stock B has volatility 40%. The two stocks are 50% correlated:

$$\sigma_A = 30\%$$

$$\sigma_B = 40\%$$

$$\rho_{AB} = 0.5$$

This two-stock universe contains a benchmark, $\mathbf{h}_B$, and a managed portfolio, $\mathbf{h}_P$:

$$\mathbf{h}_B = \begin{bmatrix} 50\% \\ 50\% \end{bmatrix}, \quad \mathbf{h}_P = \begin{bmatrix} 55\% \\ 45\% \end{bmatrix}$$

- a) What is the covariance matrix, $\mathbf{V}$?
- b) What is the volatility, σ_P , of the managed portfolio?
- c) What is the active risk, ψ_P , of the managed portfolio?
- d) We denote the vector of 1's as $\mathbf{e}$. In this problem with 2 assets:

$$\mathbf{e} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Calculate the vector $\mathbf{V}^{-1} \cdot \mathbf{e}$, and the number $\mathbf{e}^T \cdot \mathbf{V}^{-1} \cdot \mathbf{e}$. What is the volatility, σ_C , of the minimum variance, fully-invested portfolio:

$$\mathbf{h}_C = \frac{\mathbf{V}^{-1} \cdot \mathbf{e}}{\mathbf{e}^T \cdot \mathbf{V}^{-1} \cdot \mathbf{e}}$$

2. (10 points) Assuming that active returns are uncorrelated, what would you estimate as the minimum active risk achievable with a 20 stock portfolio relative to the S&P 500? Assume that each stock has active risk of 25%, and that the S&P 500 is an equal weighted benchmark. Would your answer increase or decrease if you correctly accounted for the true capitalization-weighting of the index?
3. (10 points) Fischer runs a long-only equity strategy following 500 stocks per quarter, with an information coefficient of 0.05. Myron is considering starting up an asset allocation strategy. He appears to have an information coefficient of 0.08, and plans to invest long-short. Into how many asset classes must he invest quarterly to match Fischer's information ratio?

4. (30 points) The file “Emissions_FF_Industry” presents the emissions intensity (emissions/total output) for most of the Fama-French 49 industries. Some are missing because the data come from government sources and their scheme differs from that used for listed equities. The file “FF_49_returns” contains monthly industry and factor returns.
- Summarize performance of the 5 highest and lowest emitting industries. Use the monthly value weighted portfolios and start in January 1970 to avoid missing data
 - Form an equal weighted portfolio of the 5 highest and 5 lowest emitting industries. Plot the cumulative returns of each on their own and of a “green” strategy going long the lowest and short the highest emitting.
 - Estimate and analyze the loadings of the high, low, and long-short portfolios. Just use the three factors (market, size, value) but if you are curious you can also add the momentum and conservative-minus-aggressive factors.
 - Present your results for the full sample, for the period from 2010 to the present, and finally for the past 13 months.
 - What has happened to the loading on the value factor HML? Why?
5. (30 points) Preparing for the Final project:
Using your results from **(a)–(c)** (top/bottom-5 industries by emissions intensity, cumulative return plots, and factor loadings across full sample, 2010–present, and past 13 months):
- Context you provide to ChatGPT (2–3 bullet summary):
Before you ask ChatGPT for ideas, summarize the key empirical facts you found.
Include:
 - The five highest and five lowest emitting industries (tickers/names) and whether ranks are stable over time.
 - Performance highlights (CAGR, annualized vol, Sharpe) for High, Low, and Green (Low–High) portfolios; mention any subperiod reversals/regime changes.
 - Factor results: notable HML/SMB/MKT loadings for each portfolio in (i) full sample, (ii) 2010–present, (iii) last 13 months; call out any HML drift and your tentative hypothesis.

(This context should be pasted at the top of your ChatGPT prompt.)
 - ChatGPT Research Step (ask for two strategies):
Ask ChatGPT to propose two implementable active strategies that explicitly leverage your emissions findings. Your prompt must require:
 - A portfolio construction method (universe, ranking, rebalancing, weighting, hedging) that builds on Low vs. High emissions signals.
 - Risk controls tied to your results (e.g., cap active sector bets, target volatility, beta-neutral or HML-neutral overlays if your regressions show unwanted exposure).

- A robustness plan across the three horizons you tested (full / 2010–present / last 13 months), plus turnover and transaction cost considerations.
- An explanation for any HML loading shifts and how the strategy will isolate the emissions effect versus value/size/market.

c) Evaluation Step:

Turn in:

- (i) Exact prompt(s) used (copy/paste).
- (ii) Your summary of ChatGPT's two strategies (5–8 bullets total).
- (iii) Plausibility check vs. your HW results:
 - i. Do the suggested constraints address the observed factor loadings (esp. HML changes)?
 - ii. Are sector and concentration limits appropriate given your industry composition?
 - iii. Do the proposed rebalancing/turnover assumptions make sense vs. your signal stability?
 - iv. Are robustness checks aligned with your three horizons and any recent regime shifts?

d) Forward Link to Final Project (2–3 sentences):

Select one of the two strategies as a candidate for your final project and state:

- What you will test next (specific metrics/diagnostics beyond HW01, e.g., HML-neutral variant, beta-neutral variant, cost-aware backtest).
- The key risk you must mitigate (e.g., HML drift, sector concentration, turnover).